

A Creative Approach to Visualizing Complex Networks for Non-Linear Storytelling

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Computer Science

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Creative Topic of Contest 2025: Dark

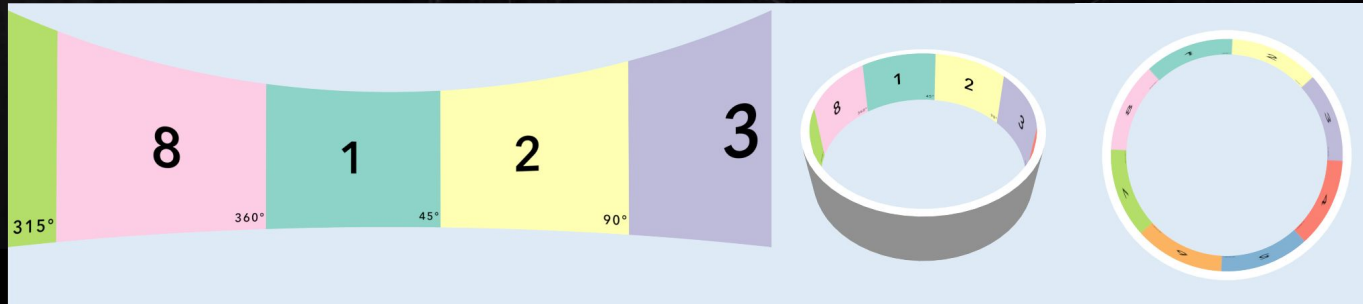


Norrköpings Decision Arena

Hardware: cylindrical screen, The room consists of a cylindrical screen with 6 meters in diameter and nine projectors, can easily be moved back and forth on the screen.

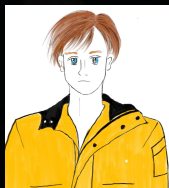
Poster: static, resolution of 12288 x 1200 pixels (This is a 10.24:1 aspect ratio, which is suitable for the 360-degree screen).

Software: the preview system, three.js based, preview poster in a 360 degree setting

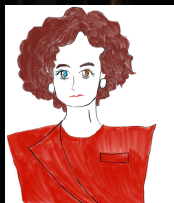


Datasets: Dark (TV series - Netflix) (1/2)

Kahnwalds family



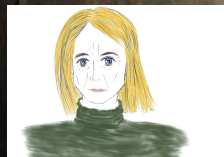
Doppler family



Nielsen family



Tiedemann Family



Mikkel was missing

Datasets: Dark (TV series - Netflix) (2/2)

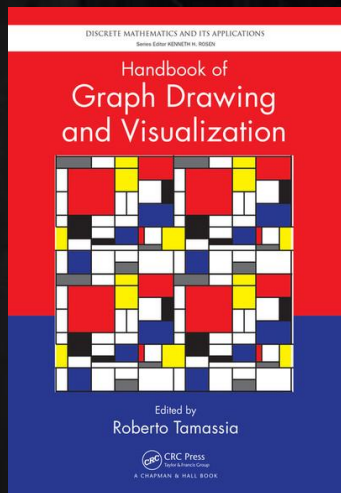
Two CSV datasets represent key narrative and structural elements of the German television series Dark.

545 nodes and 627 edges

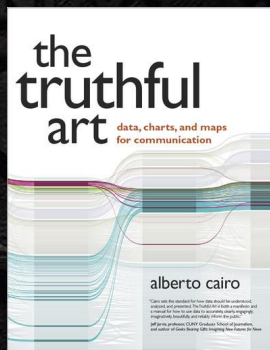
D A R K

- Dark Edges.csv includes: ID, Source, Target, Type, and Description.
- Dark Events.csv includes: ID, World, Date, Important Trigger, Death, Description, and Characters.

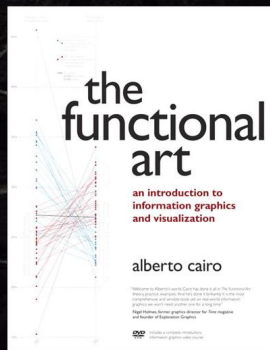
Literature



This book helped me familiarize myself with various layout algorithms, get a general sense of graph drawing as a **beginner**.



What made difference between the **IDP** and the **final result** of the Bachelor Thesis.



Through these books on information design, I discovered D3.js and realized that creativity in this field is boundless.

Tools



Inkscape



Visual Studio Code



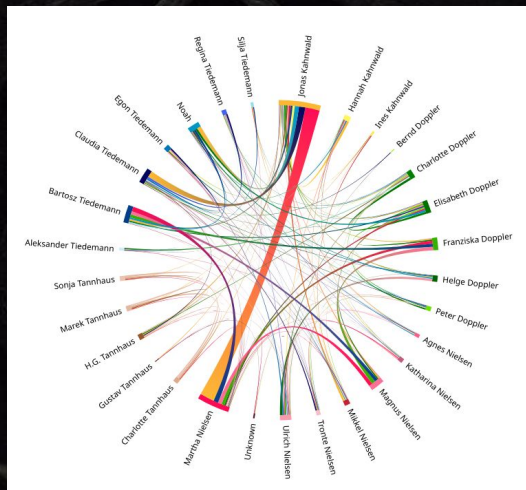
Five Graphs (1/4)

Process:

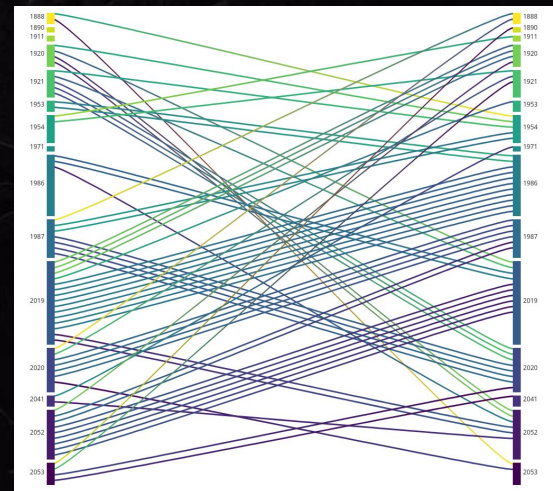
1. Prepare the data using python or LibreOffice Calc
2. Create the graphs with **D3.js**
3. Apply style, colors, and decorative design elements
4. Narrative, the 5 graphs are connected for storytelling



- Linear Gradients
- Viridis palette



Chord graph: Characters' interactions

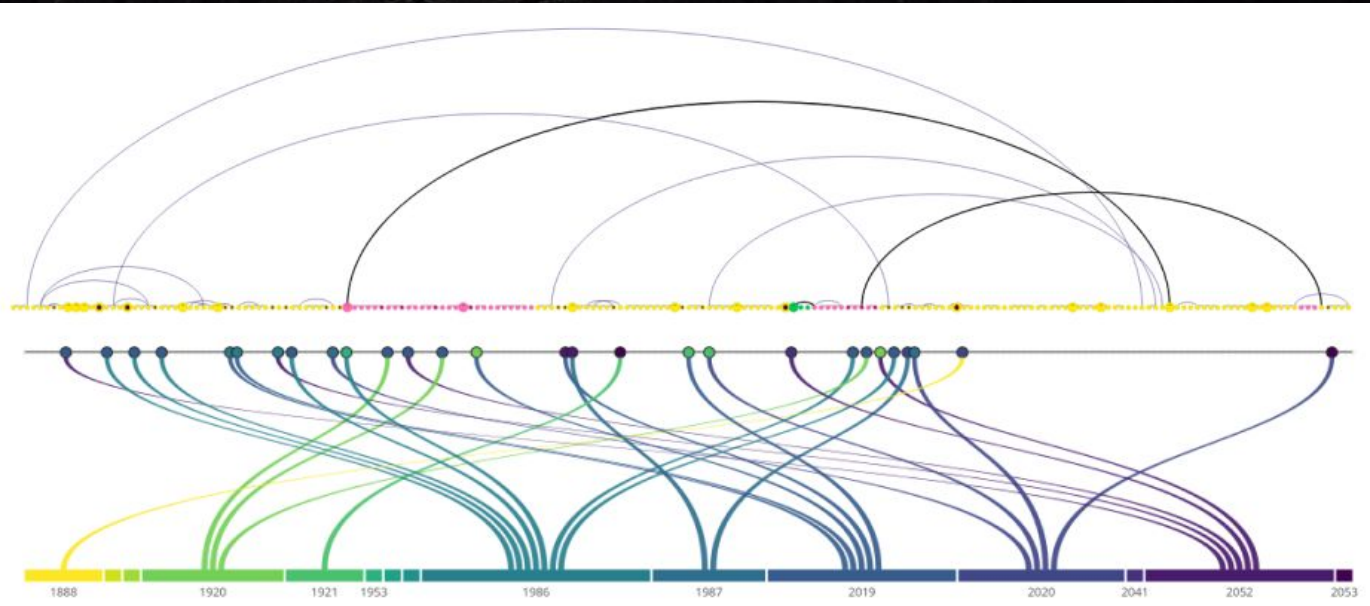


Two-Axis Alluvial Diagram

Five Graphs (3/4)

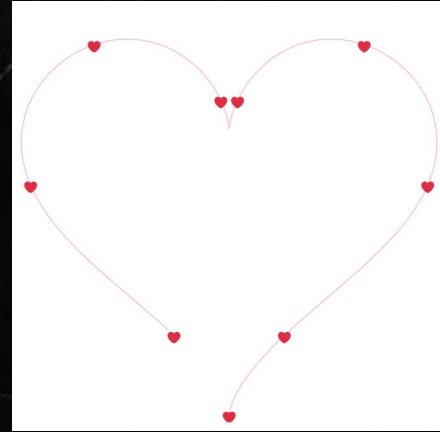
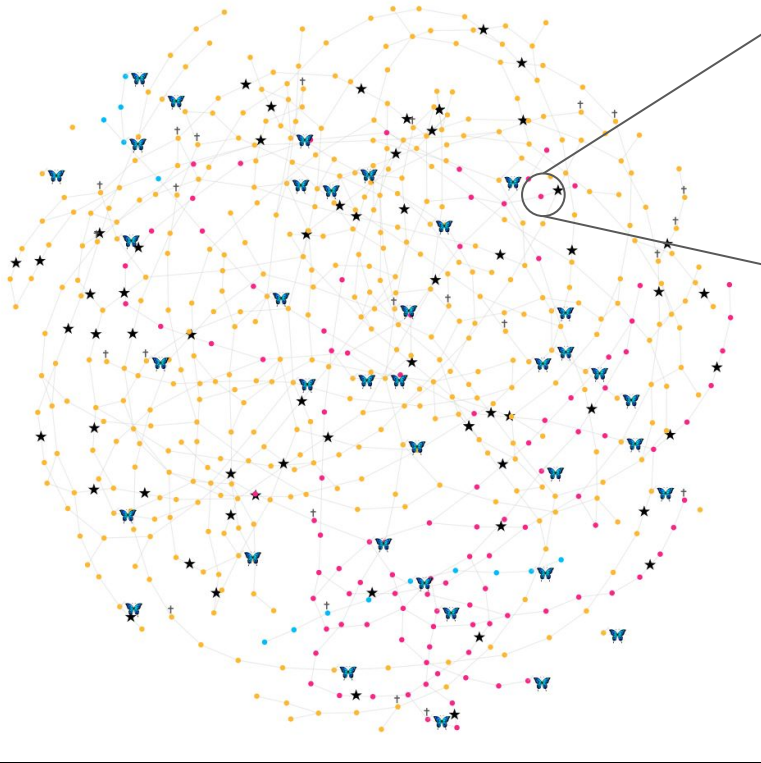
Main Graph

Arc Diagram



Alluvial Diagram

Five Graphs (4/4)



Heart Graph

The Planet: Force-Directed Graph

The insight of dark story: Love

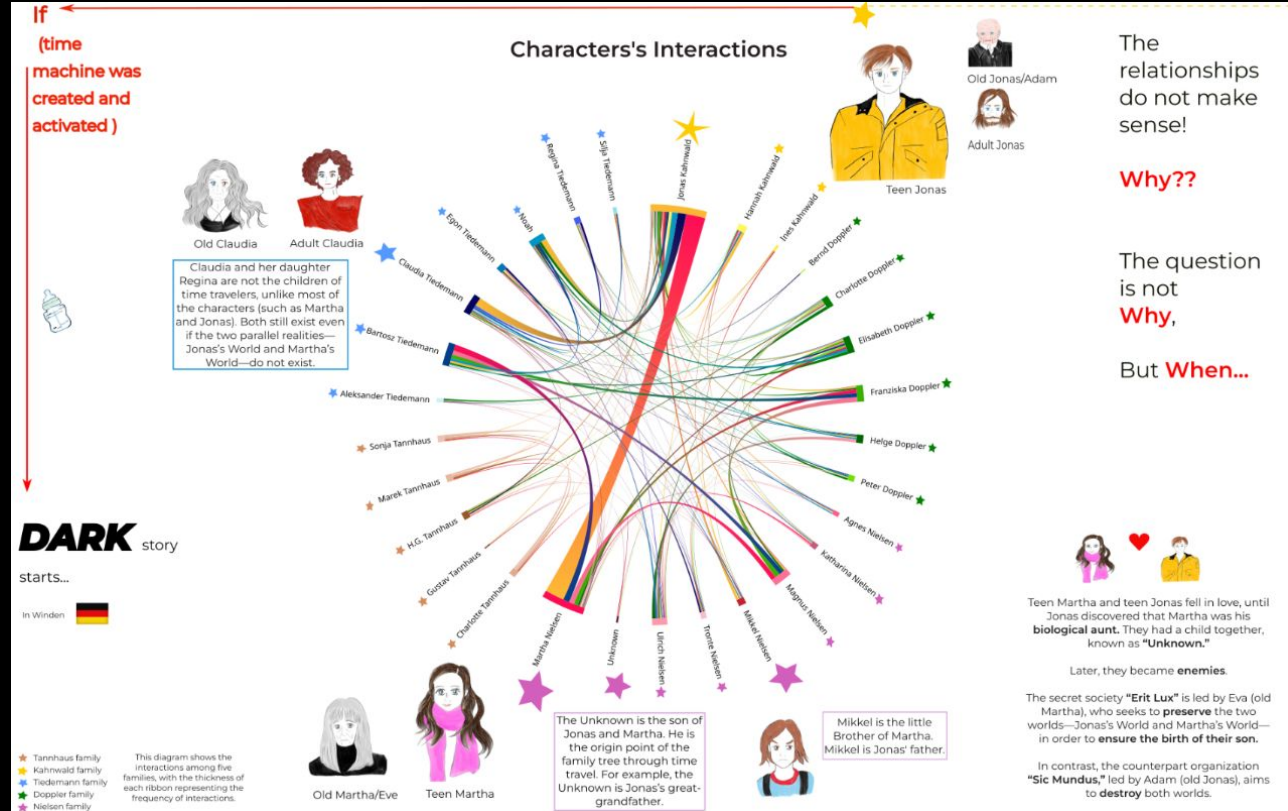
Decorations (1/2)



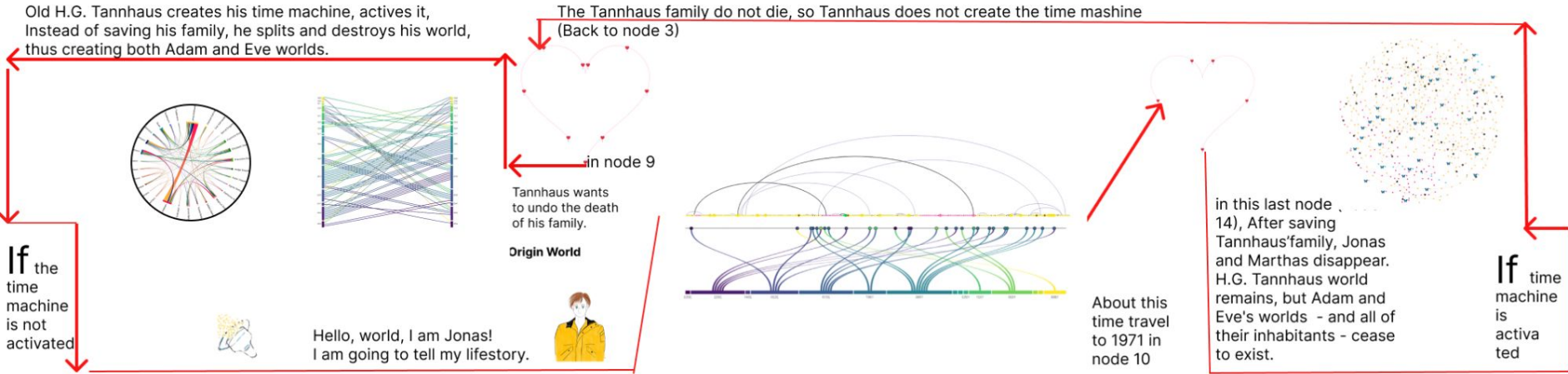
Decorations (2/2)



The Starry Night
by Vincent van Gogh



The structure: Everything is connected



- There are two “IF loops”
- The first heart graph connects to the first “IF loop” at the beginning
- The second heart graph is connected to Jonas’s timeline graph and the second “IF loop”